

Review of Battery Cell Balancing Techniques

Jian Qi

School of Electrical and Information Engineering,
The University of Sydney,
NSW 2006, Australia
jian.qi@sydney.edu.au

Dylan Dah-Chuan Lu

School of Electrical and Information Engineering,
The University of Sydney,
Sydney, Australia
dylan.lu@sydney.edu.au

Abstract—A highly reliable and efficient battery management system (BMS) is crucial for applications that are powered by electrochemical power. Cell balancing is one of the most important features of a BMS. Cell balancing techniques help to distribute energy evenly among battery cells. Without cell balancing, a portion of the capacity or energy in the battery bank will be wasted, especially for long battery string which operates in frequent recycling condition. In this paper, some popular cell balancing techniques are described and categorized according to the way of processing redundant energy in battery cells.

Index Terms—Battery; Cell Balancing; Review

I. INTRODUCTION

IN order to meet energy storage requirements, rechargeable batteries have been widely adopted in various areas, such as renewable energy storage systems, electric vehicles and telecommunication industry. In most of the applications, due to the low terminal voltage of individual battery cells, they are normally connected in series to form a battery bank to reach required voltage level.

However, the imbalance between individual battery cells in a battery bank is a common phenomenon. A well-known imbalance is State-of-Charge (SoC) differences between cells [1]. This imbalance is caused by both intrinsic and extrinsic differences among battery cells [2]. Manufacture processing variation is the main contributor of intrinsic differences. Fabricating two cells with exact same characteristics is highly unlikely. Different capacities, self-discharge rates and internal resistances lead to variation of cell performance in operation. Extrinsic factors include temperature impact and external circuitry. Uneven temperature distribution in battery bank affects the cell characteristics and lead to performance variation [3]. Higher operating temperature improves capacity and power but it accelerates aging of battery cells. Lower operating temperature degrades the performance. The other extrinsic difference is external circuitry attached to each cell that is used for sensing, control and protection of the cells. These circuitries consume different power from each battery cell and will worsen imbalance among battery cells.

The most significant impact of cell imbalance is energy loss in the battery bank. Due to the aforementioned differences between cells, energy consumption of each cell is different. This causes the SoC differences among cells. Cells with lowest and highest SoC will limit the whole battery bank performance.

For example, a Lithium-ion battery bank would have safety issue if overcharging or over-discharging happens. During discharging, the cell with lowest SoC will reach the lower limit of safe operation voltage first thus the BMS has to turn off the operation. Similar situation happens during charging, the cell with highest SoC will reach the upper limit of safe operation voltage first thus BMS has to stop the charging process. As a result, part of the energy and capacity in the other cells is wasted. Fig.1 illustrates the energy loss and capacity loss in unbalanced situation.

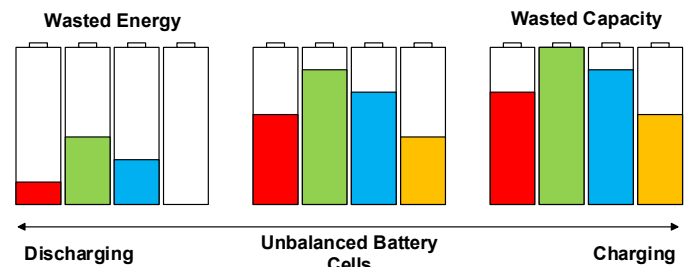


Figure 1, energy loss and capacity loss under unbalanced situation

In order to maintain a balanced battery bank, many possible topologies have been proposed [4]. All of the topologies are working under three basic approaches as shown in Fig. 2. The first approach is releasing redundant energy from battery cell. The second approach is transferring energy from fuller cells to emptier cells. The last approach is controlling the amount of charging/discharging current of each cell so that the SoC of all cells are kept at the same level.

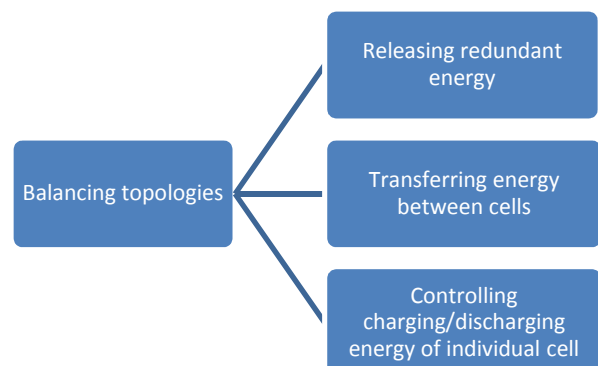


Figure 2, Cell Balancing Topologies Category

The topologies described in this paper are organized based on these three approaches. In section II, a method working on dissipating energy is explained. Topologies that transfer energy between cells are presented in section III. Section IV introduces the method using converters to control charging/discharging energy for individual battery cells. Conclusion is given in section V.

II. RELEASING REDUNDANT ENERGY

Dissipating energy from fuller cells is a reliable and simple cell balancing approach. However, this approach is inefficient. Releasing redundant energy is only designed to balance battery cells during charging. It is meaningless to perform this balancing method when battery is discharging because no energy can be saved.

Switched shunt resistor method [4-6] supports balancing during charging process. Fig.3 shows a typical circuit diagram of switched shunt resistor. As shown in the schematic, each cell is associated with a switch and a resistor in parallel. Two modes of operation can be applied to this topology. The first mode has a simple control. All switches are only turned on during charging process. If one cell has higher voltage level, more current will go through the resistor instead of charging the cell and vice versa. As a result, emptier cells will catch up fuller cells effectively if the resistance is properly designed. The second mode involves voltage sensing and complex switch control. The voltages of each cell are sensed. Switches are turned on and off according to the voltage levels. The fuller cells will dissipate more energy through the resistor.

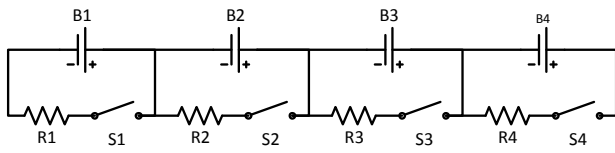


Figure 3, Switched Shunt Resistor

III. TRANSFERRING ENERGY BETWEEN CELLS

Compared to first approach, transferring energy from fuller cells to emptier cells instead of simply dissipating redundant energy is more reasonable and efficient. It can be applied during both charging and discharging. An intermediate bridge which can absorb energy from fuller cell and deliver to emptier cell is used. This bridge can be capacitors, inductors, transformers or switching converters.

A. Switched Capacitors

Switched capacitors method [7] uses capacitors as the bridge. The basic circuit diagram is shown in Fig.4. The working principle is to synchronize the voltages of two adjacent battery cells by connecting a common capacitor in parallel alternatively. As shown in the schematic, all of the switches will be at position 1 or position 2 at the same time. The redundant energy of the fuller cell in two adjacent cells will be stored in the capacitor. When switches change position, the energy will then be transferred to the emptier cells. In the long

run, all the cells will be balanced gradually. The control strategy of the switches is very simple. All of the switches can be controlled by the same control signal with a fix duty cycle. 45% duty cycle for position 1 and position 2 is a common setting. The 10% duty cycle for disconnecting capacitor from battery cells is to prevent shoot-through failure.

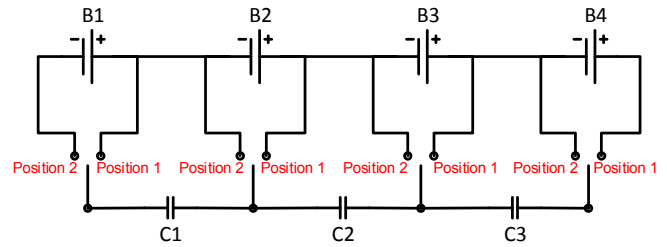


Figure 4, Switched Shunt Capacitor

A significant advantage of this method is avoiding floating voltage sensing, closed-loop control and calculations of SoC. This method is so called automatic balancing method. However, aging problem of the battery cells would degrade effectiveness of voltage balancing due to the variation in capacities among cells. Another drawback of this method is low balancing speed when the cell string is very long. This method can only allow energy transferring between adjacent cells. For example, the energy in B1 has to go through all the cells in between to charge B4. In order to improve the cell balancing speed, Double-Tiered Switched-Capacitor [8, 9] and Chain-Structure Switched-Capacitor [10] are proposed. The goal of these proposed topologies is focused on reducing the number of 'bypassing' battery cells to increase the cell balancing speed.

B. Double-Tiered Switched-Capacitor

Double-Tiered Switched-Capacitor is constructed by adding a new tier of capacitors in parallel with the original switched capacitors. The schematic diagram is shown in Fig.5. Capacitors C4 and C5 are added to provide additional paths for energy transferring. Battery cells that are not adjacent can transfer energy through the additional capacitors. For example, energy in cell B1 can be transferred to B3 in one switching cycle through C4 instead of going through cell B2 in two switching cycles. Hence, balancing speed is significantly improved.

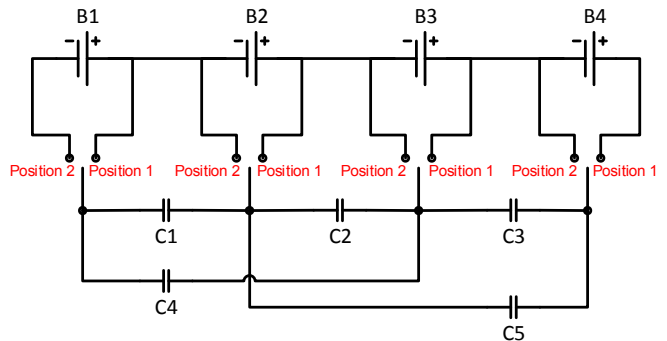


Figure 5, Double-Tiered Switched-Capacitor

C. Chain-Structure Switched-Capacitor

Chain-Structure Switched-Capacitor has an additional path between the first cell and the last cell in a battery string, which forms a circle path. By having the additional path, the number of ‘bypassing’ battery cells between the first and the last cells is halved compared to the original arrangement. The schematic diagram is shown in Fig.6. Two additional double pole single throw (DPST) switches S1 and S2 and one capacitor C4 are added. C4 is used to balance B1 and B4 and it is connected by S1 and S2 alternatively.

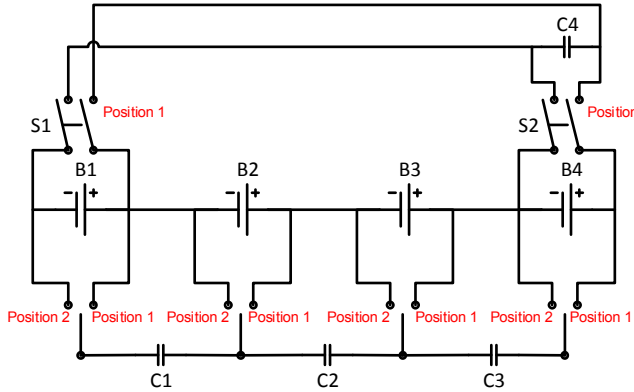


Figure 6, Chain-Structure Switched-Capacitor

D. Transformer Balancing

Transformer is another reliable candidate of the intermediate bridge. Transformer balancing methods [4, 6, 11-14] transfer energy from one battery cell to another cell by using different types of transformer. Multi-winding transformer method and multiple transformer method are proposed. In these two topologies, only one switch is used and it does not require closed-loop control. The working principles of these two methods are the same.

Multi-winding transformer method (shown in Fig.7) has a single core with one primary winding and several secondary windings. Primary winding is connected in parallel with the battery bank. Each secondary winding is associated with a diode rectifier and connected with a battery cell. The reactance of the secondary windings is directly related to the associated battery cell voltage level. Higher voltage level will have larger reactance. The current induced in secondary will be smaller if battery cell voltage is higher. In other words, the charging current of each cells is inversely proportional to its voltage level (voltage level is proportional to SoC). As such, emptier cells will catch up with the fuller cells. However, this method suffers from complex magnetic design with increasing number of windings and the limitation of the number of secondary windings per transformer.

Multiple transformer method (shown in Fig.8) uses several transformers instead of one. All the primary windings are connected in parallel and all the secondary windings are connected to each cell with a diode rectifier. The result is exactly the same as multi-winding transformer method. The number of cells is no longer limited and it is simple to connect additional cells without changing controller.

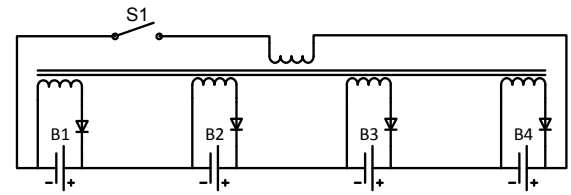


Figure 7, Multiple-Winding Transformer Balancing

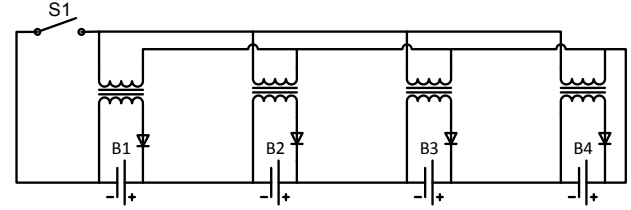


Figure 8, Multiple Transformers Balancing

E. Buck-Boost Converter Balancing

Buck-boost converter method [15] uses simplified bidirectional buck-boost as the bridge to transfer energy between two adjacent battery cells. The schematic is shown in Fig.9. Two switches (S1 and S2) and an inductor (L1) form a bidirectional buck-boost converter. The voltage levels of each battery cells are sensed and the voltage differences of two adjacent cells are calculated. The voltage differences information is the primary data to control the energy flow of each buck-boost converter. By calculating the necessary duty cycle according to the voltage differences, the energy flow of each converter can be controlled, and hence balancing is achieved.

This method involves complex voltage sensing and control. It also requires more components than the switched capacitor methods.

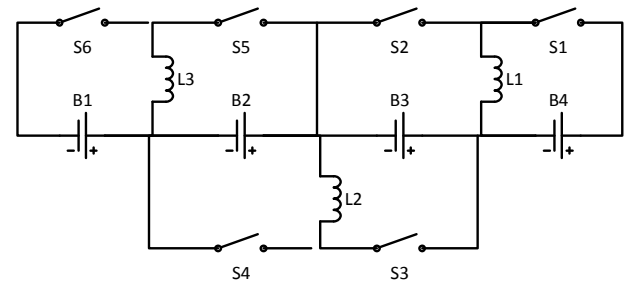


Figure 9, Buck-Boost Converter Balancing

F. Cuk Converter Balancing

Other bidirectional converter like Cuk converter can be used to perform cell balancing as well [16, 17]. The basic arrangement and control strategy are exactly the same as that of the buck-boost converter balancing method. A bidirectional Cuk converter connects two adjacent cells as shown in Fig.10. The cell voltage differences determine the control of the switches so as to control the energy flow between two adjacent cells.

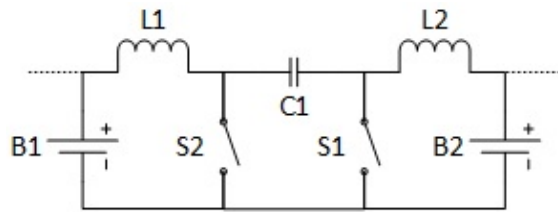


Figure 10, Cuk Converter Cell Balancing

Similar to the switched capacitor method, redundant energy in fuller cell cannot be directly transferred to emptier cell for aforementioned converter balancing methods. The energy has to be flowing through the cells in between the strongest cell and the weakest cell. This will lead to slow balancing speed.

G. Switch Matrix Balancing

Both switched capacitor balancing and basic converter balancing approaches have the same natural shortage which is they can only transfer energy between adjacent battery cells. Switch matrix balancing method [4, 18-20] provides a way to selectively balance any battery cell in the battery bank. Switch matrix balancing method consists of an array of switches and a DC/DC converter as shown in Fig. 11.

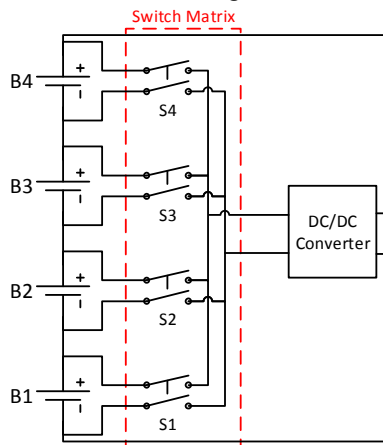


Figure 11, Switch Matrix with DC/DC Converter Balancing

As shown in Fig. 11, each battery cell is connected with a DPST switch which is used to connect the cell to the DC/DC converter for charging. The DC/DC converter is a step down converter and its input is the battery bank voltage. The charging current can be controlled by performing output current feedback control. Sensing of cell voltage is essential and the cell with lowest voltage will be connected with the DC/DC converter. To avoid short circuit between battery cells, it is vital to have proper logic control. Compared to switched capacitor and basic converter balancing approaches, the balancing speed of switch matrix balancing approach is faster. Another advantage of this method is the simplicity of modularization.

Switch matrix can also be used with capacitor and transformer as shown in Fig. 12 to improve balancing speed and to reduce the number of transformer and windings. The working principle is similar to the switch matrix with DC/DC converter.

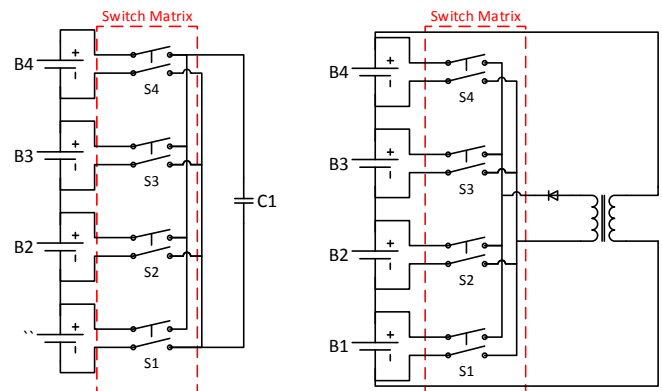


Figure 12, Switch Matrix with Capacitor and Transformer

H. Multiple Input Fly-Back Converter Balancing

A topology based on fly-back converter [21] as shown in Fig. 13 is proposed. This topology is similar to multi-winding transformer (performs like a unidirectional fly-back converter) method where the diode rectifiers are replaced by switches. By doing so, it becomes a bidirectional converter. Two balancing strategies can be developed. Energy in individual cells can be transferred to the battery bank (called top balancing), and energy of the battery bank can charge individual cells (called bottom balancing). Top balancing is applied during charging process to prevent overcharging of individual cells. Bottom balancing is applied during discharging process to prevent overdischarging.

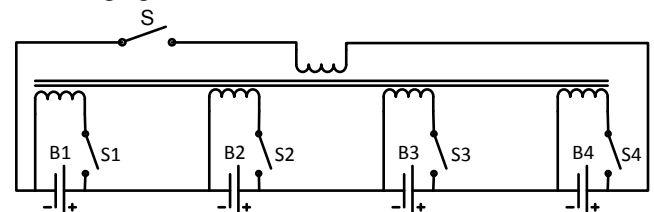


Figure 13, Fly-Back Converter Balancing

IV. CONTROLLING CHARGING/DISCHARGING ENERGY OF INDIVIDUAL CELL

Another approach to achieve cell balancing is to control the energy of individual cell delivered to the load. No energy is transferred between cells and no energy is dissipated through resistors. Multiple converters or multilevel converters are suitable candidates for this approach. One significant advantage of this approach is the optimum management of individual cell.

A. Using a Single Converter for Every Cell

Using a single power converter for every cell [22, 23] is a reasonable solution to control the power flow of each cell during operation. Fig. 13 shows the structure of this method. The attached boost converters provide an additional current path for each cell. The total current draw from individual cell can be properly controlled according to either the SoC or cell voltage. If B1 has lower SoC, less current will be absorbed by the converter so that the power delivered by B1 is lower than other cells, and vice versa. As a result, the SoC of all cells will be kept at an even level.

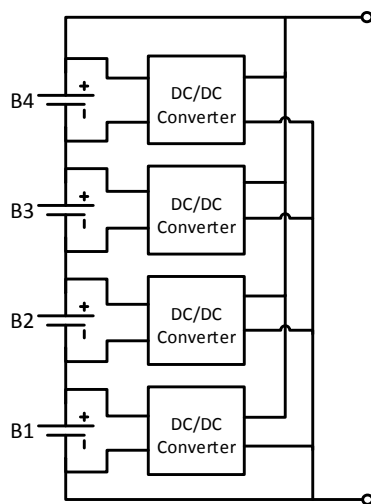


Figure 13, a single converter for each cell

B. Cascaded Multilevel Half-Bridge Converter

Multilevel converters are widely used in DC/AC inversion. They have a number of input sources. When using battery cells as the input sources, it avoids the series connection between battery cells directly. The amount of energy delivered by a cell can be separately controlled since cell balancing is easily realized. A cascaded multilevel half-bridge converter for battery management is introduced in [24] and the schematic is shown in Fig. 14. With the help of half-bridge arms, the number of connected battery cells can be controlled to produce a variable output voltage. A sinusoidal voltage can be produced with n (the number of battery cells) voltage steps. The cell charging/discharging energy is determined by the duration of this particular cell connected to the output. Cells with higher SoC can be discharged more or charged less during operation. By using proper control strategy, energy in every cell will be even.

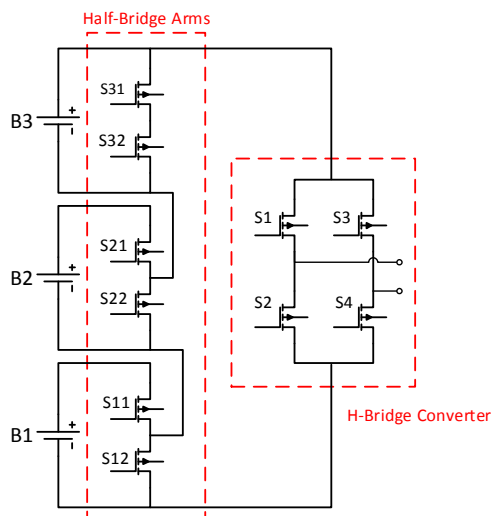


Figure 14, Cascaded Multilevel Half-Bridge Converter Balancing

The main function of these converters is to regulate output voltage. Cell balancing can be achieved by modifying control strategy. One significant advantage is that no extra circuitry is needed to balance cells. Another advantage is fault tolerance. If

any of the battery cells reaches its end-of-life or is malfunctioned, this topology can continue deliver power to the load.

V. CONCLUSION

Different techniques of battery cell balancing are described. All of the techniques can be grouped into three categories based on the way of processing redundant energy in fuller cells. The first category is based on dissipating redundant energy through resistive load. The second category is focused on transferring exceed energy in fuller cells to emptier cells. The last category is based on controlling charging/discharging energy of individual cells.

Wasting redundant energy is a huge drawback of the first approach. And this approach is not applicable in online balancing. The second approach utilizing the redundant energy efficiently. However, it requires extra circuitry and well control algorithm. Individual battery cell control is achieved in the third approach. This approach may require more complex circuitry and control method.

Advanced control of battery cells and fault tolerance of battery bank are crucial in future battery applications, especially in frequently recycling applications. Even though many cell balancing topologies have been proposed, there is still no single topology which has all the advantages, such as cost effective, high balancing speed and high reliability. Further investigations in battery cell balancing and battery cell management are required.

VI. REFERENCES

- [1] I. Aizpuru, U. Iraola, J. M. Canales, E. Unamuno, and I. Gil, "Battery pack tests to detect unbalancing effects in series connected Li-ion cells," in *Clean Electrical Power (ICCEP), 2013 International Conference on*, 2013, pp. 99-106.
- [2] K. Jonghoon, S. Jongwon, C. Changyoon, and B. H. Cho, "Stable Configuration of a Li-Ion Series Battery Pack Based on a Screening Process for Improved Voltage/SOC Balancing," *Power Electronics, IEEE Transactions on*, vol. 27, pp. 411-424, 2012.
- [3] J. R. Belt, C. D. Ho, T. J. Miller, M. A. Habib, and T. Q. Duong, "The effect of temperature on capacity and power in cycled lithium ion batteries," *Journal of Power Sources*, vol. 142, pp. 354-360, 3/24/ 2005.
- [4] J. Cao, N. Schofield, and A. Emadi, "Battery balancing methods: A comprehensive review," in *Vehicle Power and Propulsion Conference, 2008. VPPC '08. IEEE*, 2008, pp. 1-6.
- [5] B. Lindemark, "Individual cell voltage equalizers (ICE) for reliable battery performance," in *Telecommunications Energy Conference, 1991. INTELEC '91., 13th International*, 1991, pp. 196-201.
- [6] S. W. Moore and P. J. Schneider, "A review of cell equalization methods for lithium ion and lithium polymer battery systems," *SAE Publication*, pp. 01-0959, 2001.

- [7] C. Pascual and P. T. Krein, "Switched capacitor system for automatic series battery equalization," in *Applied Power Electronics Conference and Exposition, 1997. APEC '97 Conference Proceedings 1997., Twelfth Annual*, 1997, pp. 848-854 vol.2.
- [8] A. Baughman and M. Ferdowsi, "Double-tiered capacitive shuttling method for balancing series-connected batteries," in *Vehicle Power and Propulsion, 2005 IEEE Conference*, 2005, pp. 109-113.
- [9] A. C. Baughman and M. Ferdowsi, "Double-Tiered Switched-Capacitor Battery Charge Equalization Technique," *Industrial Electronics, IEEE Transactions on*, vol. 55, pp. 2277-2285, 2008.
- [10] K. Moon-Young, K. Chol-Ho, K. Jun-Ho, and M. Gun-Woo, "A Chain Structure of Switched Capacitor for Improved Cell Balancing Speed of Lithium-Ion Batteries," *Industrial Electronics, IEEE Transactions on*, vol. 61, pp. 3989-3999, 2014.
- [11] N. H. Kutkut, D. M. Divan, and D. W. Novotny, "Charge equalization for series connected battery strings," *Industry Applications, IEEE Transactions on*, vol. 31, pp. 562-568, 1995.
- [12] N. H. Kutkut, H. L. N. Wiegman, D. M. Divan, and D. W. Novotny, "Design considerations for charge equalization of an electric vehicle battery system," *Industry Applications, IEEE Transactions on*, vol. 35, pp. 28-35, 1999.
- [13] H. Sakamoto, K. Murata, K. Nishijima, K. Harada, S. Taniguchi, K. Yamasaki, *et al.*, "Balanced charging of series connected battery cells," in *Telecommunications Energy Conference, 1998. INTELEC. Twentieth International*, 1998, pp. 311-315.
- [14] L. Siqui, C. C. Mi, and Z. Mengyang, "A High-Efficiency Active Battery-Balancing Circuit Using Multiwinding Transformer," *Industry Applications, IEEE Transactions on*, vol. 49, pp. 198-207, 2013.
- [15] K. Nishijima, H. Sakamoto, and K. Harada, "A PWM controlled simple and high performance battery balancing system," in *Power Electronics Specialists Conference, 2000. PESC 00. 2000 IEEE 31st Annual*, 2000, pp. 517-520 vol.1.
- [16] Z. Zhang and S. Cuk, "A high efficiency 1.8 kW battery equalizer," in *Applied Power Electronics Conference and Exposition, 1993. APEC '93. Conference Proceedings 1993., Eighth Annual*, 1993, pp. 221-227.
- [17] Y.-S. Lee and C. Ming-Wang, "Intelligent control battery equalization for series connected lithium-ion battery strings," *Industrial Electronics, IEEE Transactions on*, vol. 52, pp. 1297-1307, 2005.
- [18] K. Chol-Ho, K. Moon-Young, and M. Gun-Woo, "A Modularized Charge Equalizer Using a Battery Monitoring IC for Series-Connected Li-Ion Battery Strings in Electric Vehicles," *Power Electronics, IEEE Transactions on*, vol. 28, pp. 3779-3787, 2013.
- [19] F. Baronti, C. Bernardeschi, L. Cassano, A. Domenici, R. Roncella, and R. Saletti, "Design and Safety Verification of a Distributed Charge Equalizer for Modular Li-ion Batteries," *Industrial Informatics, IEEE Transactions on*, vol. PP, pp. 1-1, 2014.
- [20] K. Chol-Ho, K. Moon-young, P. Hong-sun, and M. Gun-Woo, "A Modularized Two-Stage Charge Equalizer With Cell Selection Switches for Series-Connected Lithium-Ion Battery String in an HEV," *Power Electronics, IEEE Transactions on*, vol. 27, pp. 3764-3774, 2012.
- [21] M. Einhorn, W. Roessler, and J. Fleig, "Improved Performance of Serially Connected Li-Ion Batteries With Active Cell Balancing in Electric Vehicles," *Vehicular Technology, IEEE Transactions on*, vol. 60, pp. 2448-2457, 2011.
- [22] M. Einhorn, W. Guertlschmid, T. Blochberger, R. Kumpusch, R. Permann, F. V. Conte, *et al.*, "A Current Equalization Method for Serially Connected Battery Cells Using a Single Power Converter for Each Cell," *Vehicular Technology, IEEE Transactions on*, vol. 60, pp. 4227-4237, 2011.
- [23] M. Einhorn, R. Permann, C. Kral, F. V. Conte, W. Guertlschmid, T. Blochberger, *et al.*, "Current equalization of serially connected battery cells for a possible second life application," in *Vehicle Power and Propulsion Conference (VPPC), 2011 IEEE*, 2011, pp. 1-5.
- [24] Z. Zheng, K. Wang, L. Xu, and Y. Li, "A Hybrid Cascaded Multilevel Converter for Battery Energy Management Applied in Electric Vehicles," *Power Electronics, IEEE Transactions on*, vol. 29, pp. 3537-3546, 2014.